NETWORKING

NAT
Visual Guide

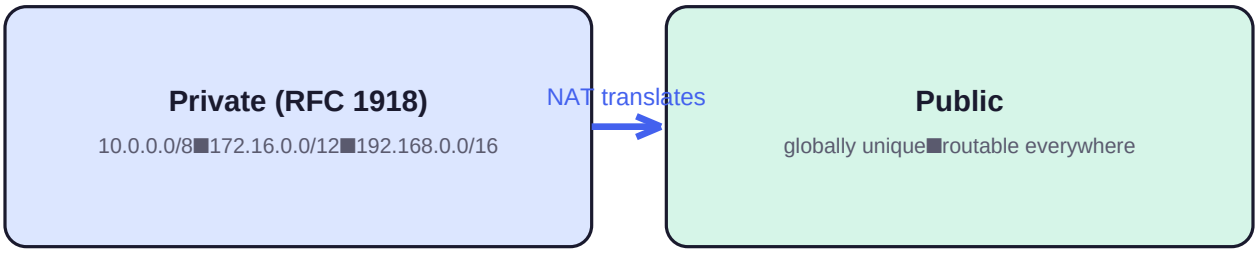
Network Address Translation —
how one public IP serves a whole network.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/5

Private vs Public

RFC 1918 carved out addresses that never route on the internet.



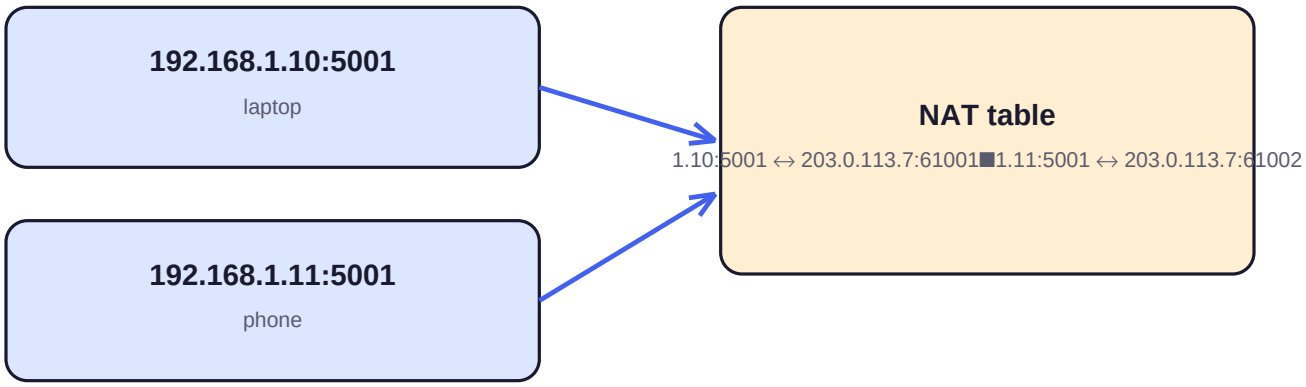
- Private addresses are reusable: every home can have 192.168.1.0/24 — they never collide because they never leave.
- NAT at the edge rewrites private → public on the way out, and reverses it on the way back.
- This is why IPv4 survived: billions of devices share a shrinking pool of public addresses.

STEP 2/5

PAT: Port Address Translation

One public IP, thousands of sessions — ports are the multiplexing key.

Inside (private) Outside (public)

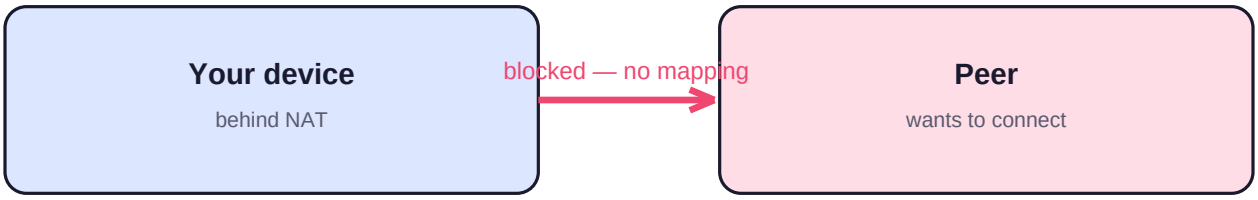


- Outbound: source IP:port rewritten to the router's public IP + a unique port.
- Inbound reply: destination port looked up in the table → translated back to the private host.
- The 5-tuple (proto, src/dst IP, src/dst port) is the session key — 65k ports per public IP per protocol.

STEP 3/5

The NAT Traversal Problem

NAT is stateful and one-directional: outsiders can't initiate in.



- Inbound packets with no matching table entry are dropped — there's nowhere to deliver them.
- Workarounds: port forwarding (static mapping), UPnP (apps request mappings), hole punching (STUN).
- Last resort: TURN relay — both peers connect OUT to a relay that bridges them (costs bandwidth).
- This is why P2P apps, VoIP, and games fight NAT — and why IPv6's end-to-end model is appealing.

STEP 4/5

CGNAT: NAT at ISP Scale

Carrier-Grade NAT — your ISP NATs you before you reach the internet.

- ISPs ran out of IPv4 too — CGNAT lets hundreds of customers share one public address (100.64.0.0/10 is the shared space).
- Double NAT (your router + CGNAT) breaks port forwarding entirely — no inbound services from home.
- Port quotas per customer are tight — heavy P2P/gaming users feel it first.

STEP 5/5

NAT Types & Behavior

Not all NATs treat new sessions the same.

Full cone

Any external host can reply once an outbound mapping exists — most permissive.

Restricted cone

Only hosts you've already contacted can send inbound.

Port-restricted

Only the exact IP:port you've contacted can reply.

Symmetric

Different external port PER destination — hardest to traverse.

STUN discovers your NAT type and public mapping — the first step every WebRTC call takes before hole punching.

STEP ✓/✓

Key Takeaways

NAT — Remember

- Private (RFC 1918) addresses never route publicly; NAT rewrites them at the edge.
- PAT multiplexes thousands of sessions onto one public IP using source ports.
- NAT is one-directional: inbound needs a mapping — hence port forwarding, STUN, TURN.
- CGNAT stacks a second NAT at the ISP; home port forwarding stops working.
- NAT types (full cone → symmetric) determine how hard traversal is.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev